

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import math
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4, suppress=True)

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Dispositivo: {device}')
```

Dispositivo: cuda

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints
NAMING_META = {
    'variante': 'topk',          # Arquitectura del SAE
    'tecnica': 'batch-topk',    # BatchTopK
    'capa': 6,                  # Layer del modelo OthelloGPT
    'juegos': 5000,             # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',
    'experiment_base_path': os.path.abspath('../../experiments'),
    'extras': {
        'expansion': 8,        # d_sae / d_in = 8192 / 512
        'k': 50,               # número de features activas L0 promedio    k
        'epochs': 1000,
        'patience': 20
    }
}

print('\nMetadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')
```

Metadata de naming:

```
variante: topk
tecnica: batch-topk
capa: 6
juegos: 5000
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
```

```
extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
```

```

) if NAMING_META.get('extras') else None

# get_plots_dir crea el directorio automáticamente
plots_dir = Path(get_plots_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
))

print(f'extras_id: {extras_id}')
print(f'plots_dir: {plots_dir}')

```

Nuevo extras registrado: ex1 → {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
extras_id: ex1
plots_dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_topk_topk\5000games\ex1\plots

3. Configuración del SAE con SAELens

Configuración usando LanguageModelSAERunnerConfig de SAELens con `architecture='batchtopk'`.

Arquitectura BatchTopK: - **BatchTopK activation:** calcula un threshold global por batch para que el **promedio** de features activas por muestra sea **k** (L0 promedio **k**, pero varía por muestra) - **Sin coeficiente de sparsidad:** la sparsity se controla directamente con **k** - **Loss = MSE únicamente** (sin penalización L1)

Parámetro	L1 (standard)	TopK	BatchTopK
Sparsity	sparsity_coef (soft)	k (hard, por muestra)	k (promedio del batch)
L0	Variable	Fijo = k por muestra	Variable por muestra, promedio k
Loss	MSE + $\cdot$ L1	MSE	MSE

Nota sobre el runner: `LanguageModelSAETrainingRunner(cfg).run()` requiere que Othello-GPT esté cargado vía `TransformerLens`. Las celdas siguientes entrenan el SAE directamente sobre los archivos `.npy` pre-extraídos usando la arquitectura `BatchTopK` definida en el config.

```

NUM_EPOCHS = NAMING_META['extras']['epochs']

cfg = {
    'architecture':      'batchtopk',
    'd_in':              512,
    'd_sae':             4096,
    'k':                 50,
    'lr':                1e-3,
    'train_batch_size_tokens': 4096,
    'hook_name':         'blocks.5.hook_resid_post',
}

cfg_summary = pd.DataFrame({
    'Parámetro SAELens': ['architecture', 'd_in', 'd_sae', 'k (L0 promedio)', 'hook_name', 'lr'],
    'Valor': [cfg['architecture'], cfg['d_in'], cfg['d_sae'], cfg['k'],
              cfg['hook_name'], cfg['lr']]
})

print('Configuración SAELens:')
print(cfg_summary.to_string(index=False))

```

Configuración SAELens:

Parámetro SAELens	Valor
architecture	batchtopk
d_in	512
d_sae	4096
k (L0 promedio)	50
hook_name	blocks.5.hook_resid_post
lr	0.001

4. Dataset de Activaciones (Train/Validation Split 80-20%)

Activaciones pre-extraídas con `sae/activations/run_extraction.py` desde la capa `blocks.5.hook_resid_post`.

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""
    def __init__(self, activations_path):
        self.activations = np.load(activations_path)
    def __len__(self):
        return len(self.activations)
    def __getitem__(self, idx):

```

```

        return torch.tensor(self.activations[idx], dtype=torch.float32)

activations_file = Path(f"../../activations/data/layer{NAMING_META['capa']}-{NAMING_META['id']}")
full_dataset = ActivationsDataset(activations_file)

# Split 80-20 reproducible
train_size = int(0.8 * len(full_dataset))
val_size    = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset, [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

train_loader = DataLoader(train_dataset, batch_size=cfg["train_batch_size_tokens"])
val_loader   = DataLoader(val_dataset,   batch_size=cfg["train_batch_size_tokens"])

dataset_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validación', 'Total'],
    'Muestras': [f'{len(train_dataset):,}', f'{len(val_dataset):,}', f'{len(full_dataset):,}'],
    'Shape':     [str(full_dataset.activations.shape), '', ''],
})
print(dataset_info.to_string(index=False))

```

```

Conjunto Muestras      Shape
Train    236,000 (295000, 512)
Validación  59,000
Total    295,000

```

5. Modelo: BatchTopK SAE

Implementación de la arquitectura BatchTopK de SAE Lens.

Diferencia clave vs ReLU + L1 y vs TopK estándar: - El encoder proyecta a `d_sae=8192` dimensiones (pre-activación) - La activación **BatchTopK** calcula un threshold global del batch (k-ésimo valor promediado); cada muestra puede tener más o menos de `k` features activas - L0 es **variable por muestra**, pero el **promedio del batch** `k` (soft constraint a nivel de batch) - No hay `sparsity_coef` ni cosine annealing - El decoder se normaliza a norma unitaria después de cada paso del optimizador

```

class BatchTopKSAE(nn.Module):
    """

```

```

Sparse Autoencoder con activación BatchTopK.
Equivalente a SAELens architecture='batchtopk'.

Encoder      : d_in -> d_sae (lineal + pre-bias centering)
Activación   : BatchTopK (mantiene el promedio de k features activas por batch)
Decoder      : d_sae -> d_in (columnas con norma unitaria)
"""
def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
    super().__init__()
    self.d_in = d_in
    self.d_sae = d_sae
    self.k = k
    self.k_aux = k_aux

    self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
    self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
    self.b_dec = nn.Parameter(torch.zeros(d_in))

    nn.init.kaiming_uniform_(self.W_enc)
    nn.init.kaiming_uniform_(self.W_dec)
    self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
    self.register_buffer('num_batches_not_active', torch.zeros(d_sae))
    self.n_batches_to_dead = 10 # latente muerto si inactivo por 10 batches seguidos

def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """Proyecta a espacio latente con activación BatchTopK"""
    x_cent = x - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    return self._topk_activation(pre_acts, threshold=threshold)

def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
    """BatchTopK: selecciona exactamente k * batch_size activaciones usando scatter."""
    if threshold is None:
        acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
        return (
            torch.zeros_like(x.flatten())
            .scatter(-1, acts_topk.indices, acts_topk.values)
            .reshape(x.shape)
        )
    return torch.where(x > threshold, x, torch.zeros_like(x))

def decode(self, hidden: torch.Tensor) -> torch.Tensor:

```

```

        return hidden @ self.W_dec + self.b_dec

def preprocess_input(self, x: torch.Tensor):
    x_mean = x.mean(dim=-1, keepdim=True)
    x = x - x_mean
    x_std = x.std(dim=-1, keepdim=True)
    x = x / (x_std + 1e-5)
    return x, x_mean, x_std

def postprocess_output(self, x_reconstruct: torch.Tensor,
                       x_mean: torch.Tensor, x_std: torch.Tensor) -> torch.Tensor:
    return x_reconstruct * x_std + x_mean

def forward(self, x: torch.Tensor, threshold=None):
    x_norm, x_mean, x_std = self.preprocess_input(x)
    x_cent = x_norm - self.b_dec
    pre_acts = F.relu(x_cent @ self.W_enc)
    hidden = self._topk_activation(pre_acts, threshold=threshold)
    recon = self.postprocess_output(self.decode(hidden), x_mean, x_std)
    return recon, hidden, pre_acts

def auxiliary_loss(self, x: torch.Tensor, x_reconstruct: torch.Tensor, acts: torch.Tensor):
    """L_aux del repo oficial: reconstruye el residual con top-k_aux dead latents."""
    dead_features = self.num_batches_not_active >= self.n_batches_to_dead
    if dead_features.sum() == 0:
        return torch.tensor(0.0, device=x.device)
    residual = x.float() - x_reconstruct.float()
    acts_topk_aux = torch.topk(
        acts[:, dead_features],
        min(self.k_aux, int(dead_features.sum().item()))),
        dim=-1,
    )
    acts_aux = torch.zeros_like(acts[:, dead_features]).scatter(
        -1, acts_topk_aux.indices, acts_topk_aux.values
    )
    x_reconstruct_aux = acts_aux @ self.W_dec[dead_features]
    return (1/32) * (x_reconstruct_aux.float() - residual.float()).pow(2).mean()

@torch.no_grad()
def update_inactive_features(self, acts):
    self.num_batches_not_active += (acts.sum(0) == 0).float()
    self.num_batches_not_active[acts.sum(0) > 0] = 0

```

```

@torch.no_grad()
def estimate_threshold(self, loader, device, n_batches: int = 10) -> float:
    """Estima un threshold estable promediando sobre múltiples batches
    (independiente del batch actual - para usar en validación/inferencia)."""
    self.eval()
    thresholds = []
    for i, batch in enumerate(loader):
        if i >= n_batches:
            break
        x = batch.to(device)
        x_norm, _, _ = self.preprocess_input(x)
        x_cent = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        mins = []
        for sample_idx in range(pre_acts.shape[0]):
            pos = pre_acts[sample_idx][pre_acts[sample_idx] > 0]
            if pos.numel() > 0:
                mins.append(pos.min().item())
        t = float(np.mean(mins)) if mins else 0.0
        thresholds.append(t)
    return float(np.mean(thresholds))

@torch.no_grad()
def make_decoder_weights_and_grad_unit_norm(self):
    """Proyecta el gradiente del decoder y renormaliza - igual que el repo oficial."""
    W_dec_normed = self.W_dec / self.W_dec.norm(dim=-1, keepdim=True)
    W_dec_grad_proj = (self.W_dec.grad * W_dec_normed).sum(-1, keepdim=True) * W_dec_normed
    self.W_dec.grad -= W_dec_grad_proj
    self.W_dec.data = W_dec_normed

# Crear modelo desde parámetros del cfg SAELens
model = BatchTopKSAE(d_in=cfg['d_in'], d_sae=cfg['d_sae'], k=cfg['k']).to(device)
num_params = sum(p.numel() for p in model.parameters())

model_info = pd.DataFrame({
    'Componente': [
        'Input', 'Encoder (W_enc)', 'Activación', 'Hidden activos', 'Decoder (W_dec)', 'Output'
    ],
    'Dimensión': [
        f'{cfg["d_in"]}',

```



```

        f'{cfg["d_in"]} -> {cfg["d_sae"]}',
        f'BatchTopK (k={cfg["k"]})',
        f'{cfg["d_sae"]} (solo {cfg["k"]} activos en promedio)',
        f'{cfg["d_sae"]} -> {cfg["d_in"]}',
        f'{cfg["d_in"]}',
        f'{model.k_aux}',
        f'{num_params:,'}
    ]
})
print(model_info.to_string(index=False))

```

Componente	Dimensión
Input	512
Encoder (W_enc)	512 -> 4096
Activación	BatchTopK (k=50)
Hidden activos	4096 (solo 50 activos en promedio)
Decoder (W_dec)	4096 -> 512
Output	512
k_aux	512
Total params	4,194,816

6. Función de Pérdida BatchTopK

Loss = MSE(reconstrucción)

- **MSE**: error de reconstrucción (igual que en L1)
- **Sin L1 penalty**: la sparsity ya está controlada por el threshold global de BatchTopK
- No hay `sparsity_coef` ni annealing — el único hiperparámetro de sparsity es `k` (promedio del batch)

```

ALPHA_AUX = 1 / 32 # peso de la pérdida auxiliar (arXiv:2412.06410)

def loss_function(x, reconstruction, acts, model):
    """
    Pérdida BatchTopK: MSE + alpha * L_aux (arXiv:2412.06410).
    - MSE      : error de reconstrucción
    - L_aux    : reconstruye el residual con las top-k_aux features para
                  evitar features muertas (alpha = 1/32)

    Args:
        x: Activaciones originales
    """

```

```

        reconstruction: Activaciones reconstruidas
        acts:           Pre-activaciones del encoder
        model:          Instancia de BatchTopKSAE
Returns:
    total_loss (scalar)
"""
mse    = F.mse_loss(reconstruction, x)
l_aux  = model.auxiliary_loss(x, reconstruction, acts)
return mse + l_aux

optimizer = torch.optim.Adam(model.parameters(), lr=cfg["lr"], betas=(0.9, 0.99))

optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning rate', 'betas', 'k (L0 promedio k)', 'k_aux', 'alpha_aux'],
    'Valor':      ['Adam', cfg["lr"], '(0.9, 0.99)', cfg["k"], model.k_aux, f'{ALPHA_AUX:.5f}'])
print(optim_info.to_string(index=False))

```

	Parámetro	Valor
	Optimizador	Adam
	Learning rate	0.001
	betas	(0.9, 0.99)
k (L0 promedio k)		50
	k_aux	512
	alpha_aux	0.03125 (1/32)

7. Loop de Entrenamiento

```

PATIENCE = NAMING_META['extras']['patience']

save_dir = Path('./saved_model')
save_dir.mkdir(exist_ok=True)
best_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='best',
    extras_id=extras_id
)

```

```

)
BEST_CKPT = save_dir / best_model_name

history = {
    'train_mse_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_mse_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
}

best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'Iniciando entrenamiento: {NUM_EPOCHS} épocas máximo')
print(f'Early stopping: patience={PATIENCE}')
print(f'k = {cfg["k"]} features activas (L0 promedio k por batch)')
print(f'k_aux = {model.k_aux} | alpha_aux = {ALPHA_AUX:.5f} (1/32)')
print(f'Best checkpoint: {BEST_CKPT}')
print('=' * 70)

for epoch in range(NUM_EPOCHS):

    # TRAIN
    model.train()
    t_mse = t_l0 = t_frac = 0.0

    for batch in train_loader:
        x = batch.to(device)
        optimizer.zero_grad()
        recon, hidden, pre_acts = model(x)
        total_loss = loss_function(x, recon, pre_acts, model)
        total_loss.backward()
        model.make_decoder_weights_and_grad_unit_norm()
        optimizer.step()
        model.update_inactive_features(hidden)

    with torch.no_grad():
        t_mse += F.mse_loss(recon, x).item()
        t_l0 += (hidden > 0).float().sum(dim=1).mean().item()

```

```

        t_frac += (hidden > 0).float().mean().item()

n = len(train_loader)
history['train_mse_loss'].append(t_mse / n)
history['train_l0_sparsity'].append(t_l0 / n)
history['train_fraction_active'].append(t_frac / n)

# VALIDATION (sin threshold - igual que training, L0 k)
model.eval()
v_mse = v_l0 = v_frac = 0.0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)
        recon, hidden, _ = model(x)
        v_mse += F.mse_loss(recon, x).item()
        v_l0 += (hidden > 0).float().sum(dim=1).mean().item()
        v_frac += (hidden > 0).float().mean().item()

nv = len(val_loader)
history['val_mse_loss'].append(v_mse / nv)
history['val_l0_sparsity'].append(v_l0 / nv)
history['val_fraction_active'].append(v_frac / nv)

print(f"Época {epoch+1:3d}/{NUM_EPOCHS} | "
      f"Train MSE: {history['train_mse_loss'][-1]:.6f} | "
      f"Val MSE: {history['val_mse_loss'][-1]:.6f} | "
      f"Train L0: {history['train_l0_sparsity'][-1]:.1f} | "
      f"Val L0: {history['val_l0_sparsity'][-1]:.1f}")

# Early stopping
if history['val_mse_loss'][-1] < best_val_mse:
    best_val_mse = history['val_mse_loss'][-1]
    best_epoch = epoch + 1
    epochs_no_improve = 0

torch.save(model.state_dict(), BEST_CKPT)
save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],

```

```

        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=cfg,
        history=history,
        optimizer=optimizer
    )
    print(f'  Mejor modelo guardado (época {best_epoch} | Val MSE: {best_val_mse:.6f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= PATIENCE:
        print(f'\nEarly stopping en época {epoch+1} (mejor: época {best_epoch})')
        break

# Restaurar mejor modelo
model.load_state_dict(torch.load(BEST_CKPT, weights_only=True))
print(f'\nMejor época: {best_epoch} | Mejor Val MSE: {best_val_mse:.6f}')

# Estimar threshold de inferencia al final del entrenamiento
inference_threshold = model.estimate_threshold(train_loader, device, n_batches=10)
print(f'Threshold de inferencia estimado: {inference_threshold:.4f}')

```

Iniciando entrenamiento: 1000 épocas máximo

Early stopping: patience=20

k = 50 features activas (L0 promedio k por batch)

k_aux = 512 | alpha_aux = 0.03125 (1/32)

Best checkpoint: saved_model\sae_topk_batch-topk_l6_5000g_ex1_best.pt

=====

Época 1/1000 | Train MSE: 0.512148 | Val MSE: 0.330629 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt

Mejor modelo guardado (época 1 | Val MSE: 0.330629)

Época 2/1000 | Train MSE: 0.287529 | Val MSE: 0.261441 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt

Mejor modelo guardado (época 2 | Val MSE: 0.261441)

Época 3/1000 | Train MSE: 0.247080 | Val MSE: 0.238306 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt

Mejor modelo guardado (época 3 | Val MSE: 0.238306)

Época 4/1000 | Train MSE: 0.228991 | Val MSE: 0.225421 | Train L0: 50.0 | Val L0: 50.0

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 4 | Val MSE: 0.225421)
Época 5/1000 | Train MSE: 0.217541 | Val MSE: 0.216588 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 5 | Val MSE: 0.216588)
Época 6/1000 | Train MSE: 0.208539 | Val MSE: 0.209532 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 6 | Val MSE: 0.209532)
Época 7/1000 | Train MSE: 0.201126 | Val MSE: 0.203451 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 7 | Val MSE: 0.203451)
Época 8/1000 | Train MSE: 0.194488 | Val MSE: 0.198740 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 8 | Val MSE: 0.198740)
Época 9/1000 | Train MSE: 0.188879 | Val MSE: 0.194748 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 9 | Val MSE: 0.194748)
Época 10/1000 | Train MSE: 0.184274 | Val MSE: 0.191795 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 10 | Val MSE: 0.191795)
Época 11/1000 | Train MSE: 0.180674 | Val MSE: 0.189290 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 11 | Val MSE: 0.189290)
Época 12/1000 | Train MSE: 0.177617 | Val MSE: 0.187440 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 12 | Val MSE: 0.187440)
Época 13/1000 | Train MSE: 0.175146 | Val MSE: 0.185978 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 13 | Val MSE: 0.185978)
Época 14/1000 | Train MSE: 0.172851 | Val MSE: 0.184532 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
Mejor modelo guardado (época 14 | Val MSE: 0.184532)

Época 15/1000 | Train MSE: 0.170976 | Val MSE: 0.183571 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 15 | Val MSE: 0.183571)
 Época 16/1000 | Train MSE: 0.169394 | Val MSE: 0.182633 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 16 | Val MSE: 0.182633)
 Época 17/1000 | Train MSE: 0.168043 | Val MSE: 0.182011 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 17 | Val MSE: 0.182011)
 Época 18/1000 | Train MSE: 0.166826 | Val MSE: 0.181256 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 18 | Val MSE: 0.181256)
 Época 19/1000 | Train MSE: 0.165682 | Val MSE: 0.180691 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 19 | Val MSE: 0.180691)
 Época 20/1000 | Train MSE: 0.164943 | Val MSE: 0.180319 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 20 | Val MSE: 0.180319)
 Época 21/1000 | Train MSE: 0.163940 | Val MSE: 0.179775 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 21 | Val MSE: 0.179775)
 Época 22/1000 | Train MSE: 0.163193 | Val MSE: 0.179532 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 22 | Val MSE: 0.179532)
 Época 23/1000 | Train MSE: 0.162557 | Val MSE: 0.179166 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 23 | Val MSE: 0.179166)
 Época 24/1000 | Train MSE: 0.161866 | Val MSE: 0.178781 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt
 Mejor modelo guardado (época 24 | Val MSE: 0.178781)
 Época 25/1000 | Train MSE: 0.161258 | Val MSE: 0.178548 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_16_5000g_ex1_best.pt

Mejor modelo guardado (época 25 | Val MSE: 0.178548)
 Época 26/1000 | Train MSE: 0.160759 | Val MSE: 0.178336 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 26 | Val MSE: 0.178336)
 Época 27/1000 | Train MSE: 0.160387 | Val MSE: 0.178163 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 27 | Val MSE: 0.178163)
 Época 28/1000 | Train MSE: 0.159798 | Val MSE: 0.177980 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 28 | Val MSE: 0.177980)
 Época 29/1000 | Train MSE: 0.159386 | Val MSE: 0.177855 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 29 | Val MSE: 0.177855)
 Época 30/1000 | Train MSE: 0.159024 | Val MSE: 0.177675 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 30 | Val MSE: 0.177675)
 Época 31/1000 | Train MSE: 0.158674 | Val MSE: 0.177530 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 31 | Val MSE: 0.177530)
 Época 32/1000 | Train MSE: 0.158384 | Val MSE: 0.177374 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 32 | Val MSE: 0.177374)
 Época 33/1000 | Train MSE: 0.158042 | Val MSE: 0.177291 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 33 | Val MSE: 0.177291)
 Época 34/1000 | Train MSE: 0.157760 | Val MSE: 0.177210 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 34 | Val MSE: 0.177210)
 Época 35/1000 | Train MSE: 0.157526 | Val MSE: 0.177092 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 35 | Val MSE: 0.177092)
 Época 36/1000 | Train MSE: 0.157210 | Val MSE: 0.177050 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt

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Mejor modelo guardado (época 36 | Val MSE: 0.177050)
Época 37/1000 | Train MSE: 0.156966 | Val MSE: 0.176936 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 37 | Val MSE: 0.176936)
Época 38/1000 | Train MSE: 0.156737 | Val MSE: 0.176897 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 38 | Val MSE: 0.176897)
Época 39/1000 | Train MSE: 0.156523 | Val MSE: 0.176778 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 39 | Val MSE: 0.176778)
Época 40/1000 | Train MSE: 0.156309 | Val MSE: 0.176745 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 40 | Val MSE: 0.176745)
Época 41/1000 | Train MSE: 0.156121 | Val MSE: 0.176630 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 41 | Val MSE: 0.176630)
Época 42/1000 | Train MSE: 0.155899 | Val MSE: 0.176570 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 42 | Val MSE: 0.176570)
Época 43/1000 | Train MSE: 0.155701 | Val MSE: 0.176475 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 43 | Val MSE: 0.176475)
Época 44/1000 | Train MSE: 0.155516 | Val MSE: 0.176525 | Train L0: 50.0 | Val L0: 50.0
Época 45/1000 | Train MSE: 0.155346 | Val MSE: 0.176322 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 45 | Val MSE: 0.176322)
Época 46/1000 | Train MSE: 0.155163 | Val MSE: 0.176217 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 46 | Val MSE: 0.176217)
Época 47/1000 | Train MSE: 0.154980 | Val MSE: 0.176090 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 47 | Val MSE: 0.176090)

Época 48/1000 | Train MSE: 0.154776 | Val MSE: 0.176039 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 48 | Val MSE: 0.176039)

Época 49/1000 | Train MSE: 0.154632 | Val MSE: 0.175995 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 49 | Val MSE: 0.175995)

Época 50/1000 | Train MSE: 0.154450 | Val MSE: 0.175849 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 50 | Val MSE: 0.175849)

Época 51/1000 | Train MSE: 0.154318 | Val MSE: 0.175780 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 51 | Val MSE: 0.175780)

Época 52/1000 | Train MSE: 0.154187 | Val MSE: 0.175736 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 52 | Val MSE: 0.175736)

Época 53/1000 | Train MSE: 0.154005 | Val MSE: 0.175724 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 53 | Val MSE: 0.175724)

Época 54/1000 | Train MSE: 0.153863 | Val MSE: 0.175601 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 54 | Val MSE: 0.175601)

Época 55/1000 | Train MSE: 0.153749 | Val MSE: 0.175392 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 55 | Val MSE: 0.175392)

Época 56/1000 | Train MSE: 0.153591 | Val MSE: 0.175347 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 56 | Val MSE: 0.175347)

Época 57/1000 | Train MSE: 0.153459 | Val MSE: 0.175276 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 57 | Val MSE: 0.175276)

Época 58/1000 | Train MSE: 0.153292 | Val MSE: 0.175174 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt

Mejor modelo guardado (época 58 | Val MSE: 0.175174)
 Época 59/1000 | Train MSE: 0.153172 | Val MSE: 0.175100 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 59 | Val MSE: 0.175100)
 Época 60/1000 | Train MSE: 0.153083 | Val MSE: 0.175063 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 60 | Val MSE: 0.175063)
 Época 61/1000 | Train MSE: 0.152999 | Val MSE: 0.175055 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 61 | Val MSE: 0.175055)
 Época 62/1000 | Train MSE: 0.152908 | Val MSE: 0.174966 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 62 | Val MSE: 0.174966)
 Época 63/1000 | Train MSE: 0.152826 | Val MSE: 0.174929 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 63 | Val MSE: 0.174929)
 Época 64/1000 | Train MSE: 0.152720 | Val MSE: 0.174883 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 64 | Val MSE: 0.174883)
 Época 65/1000 | Train MSE: 0.152639 | Val MSE: 0.174919 | Train L0: 50.0 | Val L0: 50.0
 Época 66/1000 | Train MSE: 0.152569 | Val MSE: 0.174827 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 66 | Val MSE: 0.174827)
 Época 67/1000 | Train MSE: 0.152488 | Val MSE: 0.174734 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 67 | Val MSE: 0.174734)
 Época 68/1000 | Train MSE: 0.152372 | Val MSE: 0.174738 | Train L0: 50.0 | Val L0: 50.0
 Época 69/1000 | Train MSE: 0.152313 | Val MSE: 0.174702 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 69 | Val MSE: 0.174702)
 Época 70/1000 | Train MSE: 0.152258 | Val MSE: 0.174783 | Train L0: 50.0 | Val L0: 50.0
 Época 71/1000 | Train MSE: 0.152185 | Val MSE: 0.174699 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt

Mejor modelo guardado (época 71 | Val MSE: 0.174699)
 Época 72/1000 | Train MSE: 0.152091 | Val MSE: 0.174601 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 72 | Val MSE: 0.174601)
 Época 73/1000 | Train MSE: 0.151979 | Val MSE: 0.174545 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 73 | Val MSE: 0.174545)
 Época 74/1000 | Train MSE: 0.151941 | Val MSE: 0.174569 | Train L0: 50.0 | Val L0: 50.0
 Época 75/1000 | Train MSE: 0.151856 | Val MSE: 0.174522 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 75 | Val MSE: 0.174522)
 Época 76/1000 | Train MSE: 0.151800 | Val MSE: 0.174485 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 76 | Val MSE: 0.174485)
 Época 77/1000 | Train MSE: 0.151716 | Val MSE: 0.174506 | Train L0: 50.0 | Val L0: 50.0
 Época 78/1000 | Train MSE: 0.151658 | Val MSE: 0.174464 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 78 | Val MSE: 0.174464)
 Época 79/1000 | Train MSE: 0.151628 | Val MSE: 0.174420 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 79 | Val MSE: 0.174420)
 Época 80/1000 | Train MSE: 0.151557 | Val MSE: 0.174403 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 80 | Val MSE: 0.174403)
 Época 81/1000 | Train MSE: 0.151498 | Val MSE: 0.174393 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 81 | Val MSE: 0.174393)
 Época 82/1000 | Train MSE: 0.151434 | Val MSE: 0.174380 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 82 | Val MSE: 0.174380)
 Época 83/1000 | Train MSE: 0.151391 | Val MSE: 0.174477 | Train L0: 50.0 | Val L0: 50.0
 Época 84/1000 | Train MSE: 0.151332 | Val MSE: 0.174353 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt

Mejor modelo guardado (época 84 | Val MSE: 0.174353)
 Época 85/1000 | Train MSE: 0.151253 | Val MSE: 0.174731 | Train L0: 50.0 | Val L0: 50.0
 Época 86/1000 | Train MSE: 0.151232 | Val MSE: 0.174331 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 86 | Val MSE: 0.174331)
 Época 87/1000 | Train MSE: 0.151177 | Val MSE: 0.174367 | Train L0: 50.0 | Val L0: 50.0
 Época 88/1000 | Train MSE: 0.151112 | Val MSE: 0.174364 | Train L0: 50.0 | Val L0: 50.0
 Época 89/1000 | Train MSE: 0.151093 | Val MSE: 0.174419 | Train L0: 50.0 | Val L0: 50.0
 Época 90/1000 | Train MSE: 0.151007 | Val MSE: 0.174255 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 90 | Val MSE: 0.174255)
 Época 91/1000 | Train MSE: 0.150951 | Val MSE: 0.174242 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 91 | Val MSE: 0.174242)
 Época 92/1000 | Train MSE: 0.150950 | Val MSE: 0.174235 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 92 | Val MSE: 0.174235)
 Época 93/1000 | Train MSE: 0.150876 | Val MSE: 0.174353 | Train L0: 50.0 | Val L0: 50.0
 Época 94/1000 | Train MSE: 0.150844 | Val MSE: 0.174206 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 94 | Val MSE: 0.174206)
 Época 95/1000 | Train MSE: 0.150775 | Val MSE: 0.174212 | Train L0: 50.0 | Val L0: 50.0
 Época 96/1000 | Train MSE: 0.150754 | Val MSE: 0.174297 | Train L0: 50.0 | Val L0: 50.0
 Época 97/1000 | Train MSE: 0.150729 | Val MSE: 0.174178 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 97 | Val MSE: 0.174178)
 Época 98/1000 | Train MSE: 0.150659 | Val MSE: 0.174176 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 98 | Val MSE: 0.174176)
 Época 99/1000 | Train MSE: 0.150623 | Val MSE: 0.174180 | Train L0: 50.0 | Val L0: 50.0
 Época 100/1000 | Train MSE: 0.150568 | Val MSE: 0.174169 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
 Mejor modelo guardado (época 100 | Val MSE: 0.174169)
 Época 101/1000 | Train MSE: 0.150536 | Val MSE: 0.174140 | Train L0: 50.0 | Val L0: 50.0
 Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt

topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 101 | Val MSE: 0.174140)
Época 102/1000 | Train MSE: 0.150500 | Val MSE: 0.174130 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 102 | Val MSE: 0.174130)
Época 103/1000 | Train MSE: 0.150475 | Val MSE: 0.174234 | Train L0: 50.0 | Val L0: 50.0
Época 104/1000 | Train MSE: 0.150445 | Val MSE: 0.174223 | Train L0: 50.0 | Val L0: 50.0
Época 105/1000 | Train MSE: 0.150419 | Val MSE: 0.174143 | Train L0: 50.0 | Val L0: 50.0
Época 106/1000 | Train MSE: 0.150365 | Val MSE: 0.174102 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 106 | Val MSE: 0.174102)
Época 107/1000 | Train MSE: 0.150319 | Val MSE: 0.174359 | Train L0: 50.0 | Val L0: 50.0
Época 108/1000 | Train MSE: 0.150318 | Val MSE: 0.174092 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 108 | Val MSE: 0.174092)
Época 109/1000 | Train MSE: 0.150269 | Val MSE: 0.174378 | Train L0: 50.0 | Val L0: 50.0
Época 110/1000 | Train MSE: 0.150264 | Val MSE: 0.174093 | Train L0: 50.0 | Val L0: 50.0
Época 111/1000 | Train MSE: 0.150187 | Val MSE: 0.174079 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 111 | Val MSE: 0.174079)
Época 112/1000 | Train MSE: 0.150166 | Val MSE: 0.174089 | Train L0: 50.0 | Val L0: 50.0
Época 113/1000 | Train MSE: 0.150130 | Val MSE: 0.174093 | Train L0: 50.0 | Val L0: 50.0
Época 114/1000 | Train MSE: 0.150109 | Val MSE: 0.174162 | Train L0: 50.0 | Val L0: 50.0
Época 115/1000 | Train MSE: 0.150100 | Val MSE: 0.174099 | Train L0: 50.0 | Val L0: 50.0
Época 116/1000 | Train MSE: 0.150039 | Val MSE: 0.174120 | Train L0: 50.0 | Val L0: 50.0
Época 117/1000 | Train MSE: 0.150031 | Val MSE: 0.174126 | Train L0: 50.0 | Val L0: 50.0
Época 118/1000 | Train MSE: 0.149984 | Val MSE: 0.174066 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 118 | Val MSE: 0.174066)
Época 119/1000 | Train MSE: 0.149948 | Val MSE: 0.174099 | Train L0: 50.0 | Val L0: 50.0
Época 120/1000 | Train MSE: 0.149942 | Val MSE: 0.174074 | Train L0: 50.0 | Val L0: 50.0
Época 121/1000 | Train MSE: 0.149921 | Val MSE: 0.174040 | Train L0: 50.0 | Val L0: 50.0
Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
topk_topk\5000games\sae_topk_batch-topk_l6_5000g_ex1_best.pt
Mejor modelo guardado (época 121 | Val MSE: 0.174040)
Época 122/1000 | Train MSE: 0.149862 | Val MSE: 0.174100 | Train L0: 50.0 | Val L0: 50.0
Época 123/1000 | Train MSE: 0.149861 | Val MSE: 0.174067 | Train L0: 50.0 | Val L0: 50.0
Época 124/1000 | Train MSE: 0.149842 | Val MSE: 0.174174 | Train L0: 50.0 | Val L0: 50.0

Época 125/1000	Train MSE: 0.149814	Val MSE: 0.174049	Train L0: 50.0	Val L0: 50.0
Época 126/1000	Train MSE: 0.149805	Val MSE: 0.174065	Train L0: 50.0	Val L0: 50.0
Época 127/1000	Train MSE: 0.149759	Val MSE: 0.174066	Train L0: 50.0	Val L0: 50.0
Época 128/1000	Train MSE: 0.149728	Val MSE: 0.174112	Train L0: 50.0	Val L0: 50.0
Época 129/1000	Train MSE: 0.149716	Val MSE: 0.174110	Train L0: 50.0	Val L0: 50.0
Época 130/1000	Train MSE: 0.149716	Val MSE: 0.174076	Train L0: 50.0	Val L0: 50.0
Época 131/1000	Train MSE: 0.149664	Val MSE: 0.174091	Train L0: 50.0	Val L0: 50.0
Época 132/1000	Train MSE: 0.149630	Val MSE: 0.174088	Train L0: 50.0	Val L0: 50.0
Época 133/1000	Train MSE: 0.149603	Val MSE: 0.174082	Train L0: 50.0	Val L0: 50.0
Época 134/1000	Train MSE: 0.149600	Val MSE: 0.174213	Train L0: 50.0	Val L0: 50.0
Época 135/1000	Train MSE: 0.149575	Val MSE: 0.174050	Train L0: 50.0	Val L0: 50.0
Época 136/1000	Train MSE: 0.149558	Val MSE: 0.174095	Train L0: 50.0	Val L0: 50.0
Época 137/1000	Train MSE: 0.149543	Val MSE: 0.174150	Train L0: 50.0	Val L0: 50.0
Época 138/1000	Train MSE: 0.149525	Val MSE: 0.174061	Train L0: 50.0	Val L0: 50.0
Época 139/1000	Train MSE: 0.149479	Val MSE: 0.174090	Train L0: 50.0	Val L0: 50.0
Época 140/1000	Train MSE: 0.149449	Val MSE: 0.174044	Train L0: 50.0	Val L0: 50.0
Época 141/1000	Train MSE: 0.149448	Val MSE: 0.174106	Train L0: 50.0	Val L0: 50.0

Early stopping en época 141 (mejor: época 121)

Mejor época: 121 | Mejor Val MSE: 0.174040

Threshold de inferencia estimado: 0.0022

8. Guardar Modelo

```
# Guardar checkpoint final
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': {
        'architecture': cfg["architecture"],
        'd_in': cfg["d_in"],
        'd_sae': cfg["d_sae"],
        'k': cfg["k"],
        'lr': cfg["lr"],
        'hook_name': cfg["hook_name"],
    },
    'history': history,
}

final_model_name = get_model_name(
```

```

    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras_id=extras_id
)
checkpoint_path = save_dir / final_model_name
torch.save(checkpoint, checkpoint_path)
print(f'Modelo guardado: {checkpoint_path}')

save_checkpoint(
    model,
    base_path=NAMING_META['base_path'],
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final',
    extras=NAMING_META['extras'],
    config=cfg,
    history=history,
    optimizer=optimizer
)

```

Modelo guardado: saved_model\sae_topk_batch-topk_16_5000g_ex1_final.pt

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk_batch-topk_16_5000g_ex1_final.pt

WindowsPath('C:/Users/Esposa/Documents/Repos/sae-othello-gpt/layer_06/models/sae_topk_batch-topk_16_5000g_ex1_final.pt')

9. Métricas básicas del modelo

```

import json

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],

```



```

NAMING_META['capa'],
NAMING_META['juegos'],
extras_id=extras_id
)

training_metrics = {
    # Calidad de reconstrucción
    'val_mse_final': history['val_mse_loss'][-1],
    'val_mse_best': best_val_mse,
    'val_l0_final': history['val_l0_sparsity'][-1],
    'val_fraction_active_final': history['val_fraction_active'][-1],
    # Contexto del entrenamiento
    'best_epoch': best_epoch,
    'total_epochs': len(history['train_mse_loss']),
    'train_mse_final': history['train_mse_loss'][-1],
    'train_l0_final': history['train_l0_sparsity'][-1],
    # Hiperparámetros TopK
    'architecture': cfg["architecture"],
    'k': cfg["k"],
    'd_sae': cfg["d_sae"],
    'd_in': cfg["d_in"],
    'expansion_factor': cfg["d_sae"] // cfg["d_in"],
    'hook_name': cfg["hook_name"],
}

os.makedirs(metrics_dir, exist_ok=True)
metrics_path = Path(metrics_dir) / 'training_metrics.json'
with open(metrics_path, 'w') as f:
    json.dump(training_metrics, f, indent=2)

print(f'Métricas guardadas: {metrics_path}')
print(pd.DataFrame(
    list(training_metrics.items()), columns=['Métrica', 'Valor']
).to_string(index=False))

```

Métricas guardadas: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae\topk_topk\5000games\ex1\metrics\training_metrics.json

Métrica	Valor
val_mse_final	0.174106
val_mse_best	0.17404
val_l0_final	50.0
val_fraction_active_final	0.012207

best_epoch	121
total_epochs	141
train_mse_final	0.149448
train_l0_final	50.0
architecture	batchtopk
k	50
d_sae	4096
d_in	512
expansion_factor	8
hook_name	blocks.5.hook_resid_post

10. Visualización de Resultados

```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# MSE Loss
axes[0].plot(history['train_mse_loss'], label='Train', color='steelblue', alpha=0.8)
axes[0].plot(history['val_mse_loss'], label='Validación', color='tomato', alpha=0.8)
if best_epoch > 0:
    axes[0].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7,
                    label=f'Best (época {best_epoch})')
axes[0].set_title('MSE Loss (Reconstrucción)')
axes[0].set_xlabel('Época')
axes[0].set_ylabel('MSE')
axes[0].legend()
axes[0].grid(True, alpha=0.3)

# L0 sparsity (para BatchTopK el promedio del batch k, pero varía por muestra)
axes[1].plot(history['train_l0_sparsity'], label='Train', color='steelblue', alpha=0.8)
axes[1].plot(history['val_l0_sparsity'], label='Validación', color='tomato', alpha=0.8)
axes[1].axhline(cfg["k"], color='orange', linestyle='--', alpha=0.8, label=f'k={cfg["k"]} (p)')
axes[1].set_title(f'L0 Sparsity (promedio del batch k={cfg["k"]})')
axes[1].set_xlabel('Época')
axes[1].set_ylabel('Features activas (promedio)')
axes[1].legend()
axes[1].grid(True, alpha=0.3)

# Fracción activa
axes[2].plot(history['train_fraction_active'], label='Train', color='steelblue', alpha=0.8)
axes[2].plot(history['val_fraction_active'], label='Validación', color='tomato', alpha=0.8)
expected_frac = cfg["k"] / cfg["d_sae"]
```

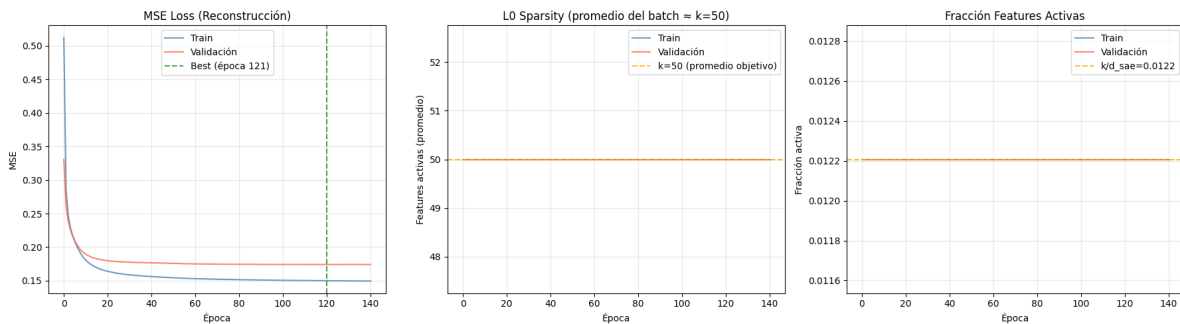
```

axes[2].axhline(expected_frac, color='orange', linestyle='--', alpha=0.8,
                label=f'k/d_sae={expected_frac:.4f}')
axes[2].set_title('Fracción Features Activas')
axes[2].set_xlabel('Época')
axes[2].set_ylabel('Fracción activa')
axes[2].legend()
axes[2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(plots_dir / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('Gráficas guardadas exitosamente')
print('='*50)

```



=====

Gráficas guardadas exitosamente

=====

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```

import subprocess

exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],

```

```

        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_batch-topk.ipynb'
output_path = exp_dir / pdf_name

print(f'Generando PDF con Quarto...')
print(f'Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f'PDF generado exitosamente: {output_path}')
else:
    print(f'Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk_topk\5000games\ex1\sae_topk_batch-topk_l6_5000g_ex1_training.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk_topk\5000games\ex1\sae_topk_batch-topk_l6_5000g_ex1_training.pdf